Global CO_2 Emissions

Trends, Regional Shifts, and Per Capita Inequalities

A data-driven overview from the Industrial Revolution to 2024

Source: Global Carbon Budget 2025 · NASA Climate Vital Signs

Atmospheric CO_2 Reaches 429 ppm — 50% Above Pre-Industrial Levels

429

parts per million

February 2026 · Mauna Loa Observatory (NOAA)

PRE-INDUSTRIAL BASELINE

~280

ppm (natural level)

Current level is >1.5× greater than natural increase at the end of the last ice age

RATE OF CHANGE

~200 yrs

vs. thousands of years

A change that took millennia now compressed into roughly two centuries

PRIMARY DRIVER

Fossil Fuel Combustion

Coal · Oil · Natural Gas

Ground-based and satellite measurements confirm the sharp acceleration

~280 ppm
Pre-industrial

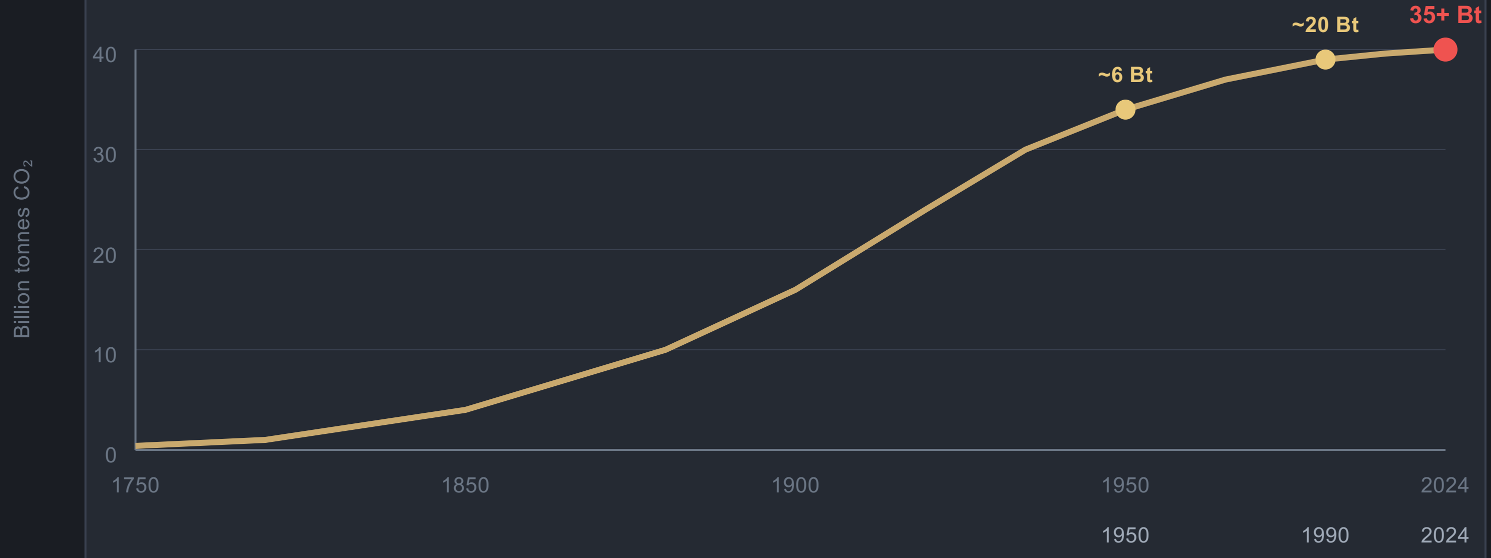
1958
Keeling starts

429 ppm
Feb 2026

Source: NOAA Mauna Loa Observatory · NASA Climate Vital Signs

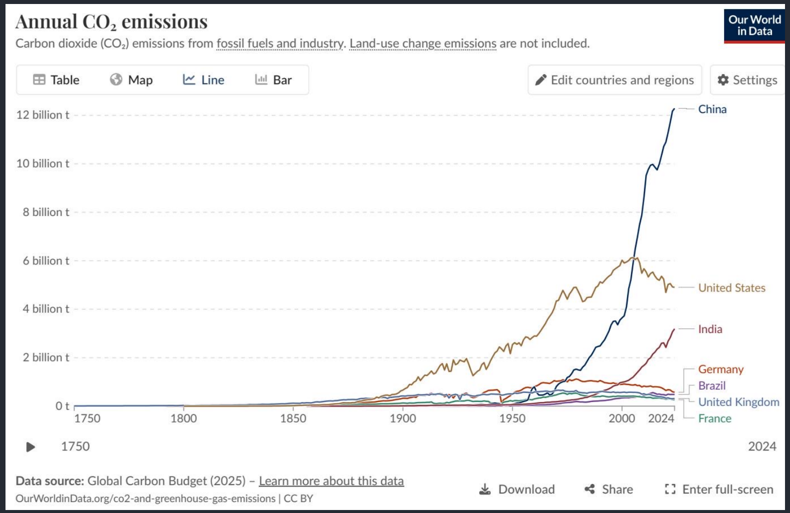
Global Fossil Fuel CO_2 Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes

GLOBAL CO_2 EMISSIONS TRAJECTORY (1750–2024)



Growth has slowed in recent years, but emissions have not yet peaked.

ANNUAL CO_2 EMISSIONS BY COUNTRY



Reference: figure_3.jpg — Country-level annual emissions showing China's steep rise post-2000 and the US decline from its ~6 Bt peak.

1950
~6 Bt
Baseline year

1990
~20 Bt
Nearly quadrupled

2024
35+ Bt
Still rising

China Now Emits Over 12 Billion Tonnes — More Than Double the United States

ANNUAL CO_2 EMISSIONS BY COUNTRY / REGION

Country / Region	CO_2 (Bt/yr)	Global Share	Trend
China	~12.5	~27%	Slight increase
United States	~5.0	~14%	Declining
India	~3.0	~8%	Rising
EU-27	~2.7	~7%	Declining
Germany	~0.6	~1.7%	Declining
United Kingdom	~0.3	~0.9%	Declining
Brazil	~0.5	~1.4%	Roughly stable
France	~0.3	~0.8%	Declining

Data: Global Carbon Budget 2025. Figures are approximate annual totals from fossil fuels and industry.

KEY INSIGHTS

China's dominance

China emits >25% of global CO_2 — more than double the US, with steep rise post-2000.

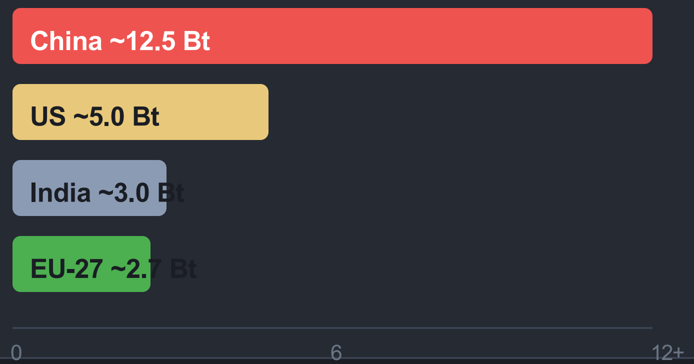
US peak and decline

The US peaked near 6 Bt around 2005 and has since declined to roughly 5 Bt annually.

India's rapid ascent

India has risen to ~3 Bt and continues to grow,

TOP 4 EMITTERS (BILLION TONNES)



Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

The geographic center of emissions has shifted dramatically over the past 124 years.

1900

>90%
Europe + United States

Rest of World

Asia, Africa, Latin America

Industrialized West dominated almost all emissions

1950

>85%
Europe + United States

Rest of World

Asia beginning to industrialize

West still overwhelmingly dominant, but share slipping

2024

~50%
Asia
Home to ~60% of population

<33%
US + Europe combined

3–4% each

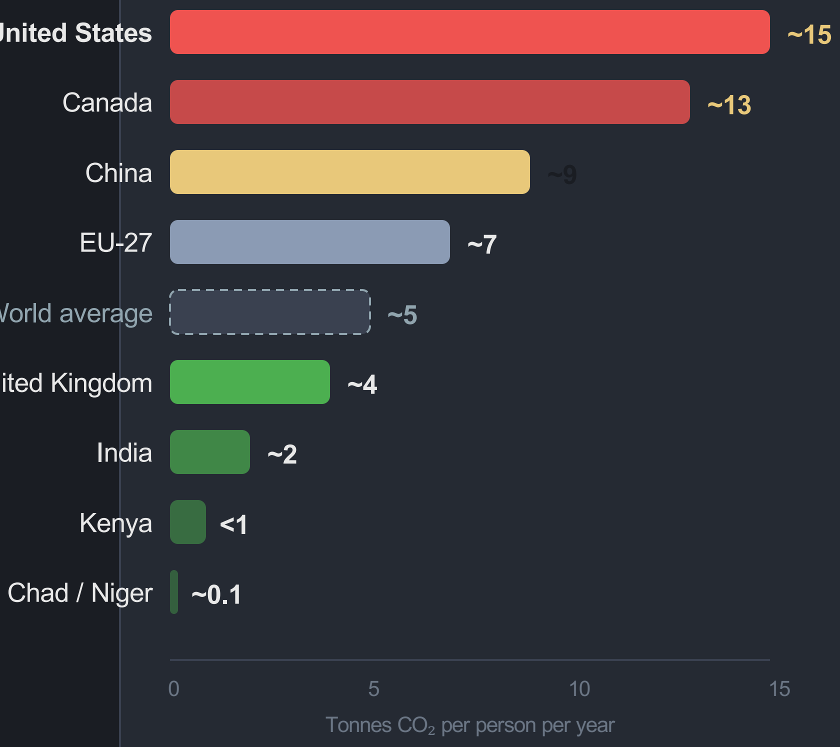
Africa & South America each comparable to aviation + shipping

The shift is structural, not cyclical.
Asia's rise reflects industrialization and population scale. Historical emitters (US/Europe) retain large cumulative responsibility despite declining shares.

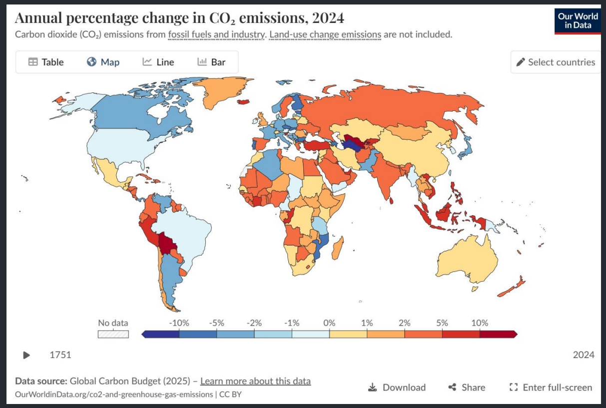
Source: Global Carbon Budget 2025 · Our World in Data

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity

PER CAPITA CO_2 EMISSIONS (TONNES / PERSON / YEAR)



PER CAPITA OVER TIME



Reference: figure_1.jpg

150×

gap between highest and lowest

The average American emits in under two days what the average person in Chad or Niger emits in an entire year.

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

Annual Percentage Change in CO_2 Emissions, 2024

Reference: figure_4.jpg — Global Carbon Budget 2025

REGIONAL PATTERNS

Declining (−1% to −5%)
United States · Much of Europe
Blue zones on map

Moderate increase (+2% to +10%)
Russia · Parts of Southeast Asia · Several African nations
Orange/red zones on map

Large increase (>+10%)
Some South American countries
Deep red zones on map

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards can have very different per capita emissions depending on energy sources.

LOWER EMISSIONS: FRANCE / UK

High nuclear share
France derives ~70% of electricity from nuclear

High and rising renewables
UK offshore wind expansion, both countries investing

Lower fossil fuel share
Reduced coal and gas dependency in power sector

PER CAPITA CO_2
~4 t / person / year

HIGHER EMISSIONS: GERMANY / NETHERLANDS

Low / declining nuclear
Germany phased out nuclear; Netherlands has minimal nuclear

Higher fossil fuel share
Continued reliance on coal (Germany) and natural gas

Moderate renewables
Growing but not yet displacing fossil baseload sufficiently

PER CAPITA CO_2
Significantly higher
than France/UK despite similar prosperity

Policy implication:
Nuclear + renewables = lower per capita emissions. Energy mix choices — not just GDP — determine national carbon intensity.

Source: Global Carbon Budget 2025 · Our World in Data

Key Takeaways

Emissions Are Still Rising, But the Map of Responsibility Is Shifting

1. **Global emissions exceed 35 Bt and have not yet peaked**
   Despite slowing growth rates, annual fossil fuel and industry CO_2 continues to rise.
   Data: Global Carbon Budget 2025

2. **China alone emits >25% of the global total (~12.5 Bt)**
   More than double the United States (~5 Bt); India (~3 Bt) is third-largest and rising fast.
   Trend: China slight increase · US declining · India rising

3. **Per capita inequalities remain extreme — a 150× gap**
   Highest emitters (US/Australia ~15 t/person) vs. lowest (Chad/Niger ~0.1 t/person).
   An American emits in <2 days what a Chadian emits in a year

4. **US and Europe are declining, but historical responsibility remains large**
   The geographic center has shifted to Asia (~50%), yet cumulative emissions matter for equity.
   1900: Europe+US >90% · 2024: Asia ~50%, US+Europe <33%

5. **Policy and energy mix choices — not just prosperity — determine outcomes**
   France vs. Germany proves similar living standards can yield very different per capita emissions.
   Nuclear + renewables = lower carbon intensity